EDA Summary: Treatment / Control Street Attributes

Datasets:

1. street_segment_qtr_attributes.csv
2. sites_db.csv

Research Question: How does parking use change after bike lanes are constructed?

1. Overview

These two files together form the treatment/control definition layer supplied by Infrastructure Victoria.

Dataset	Rows	Role
street_segment_qtr_attributes	30,783	Quarterly time-series of street-segment attributes (parking capacity, intervention flags, construction status) from ~1999 to 2025/26
sites_db	637	Site-level register of intervention and control locations, with parking capacity, disruption dates, and intervention summaries

They define which streets are treatment vs control, when interventions occurred, and how on-street parking capacity changed.

2. Dataset 1 – Quarterly Street Segment Attributes

Shape: 30,783 rows × 15 columns

Key columns

Column	Description
QUARTER	Quarter label (e.g. 2021Q2, 9900Q2) – 105 unique values spanning ~1999–2026
SiteID	Site identifier (167 unique values)
STREET_SEGMENT_ID	Street segment identifier (292 unique values)
Baseline	1 = baseline / pre-intervention observation; 0 = post; mostly null outside intervention windows

Column	Description
InterventionType	ProtectedBikeLane, Pedestrianisation, Regional Reallocation, Control, or blank
InterventionSummary	Short text description
is_under_construction / other_major_construction	Construction flags
StreetInScopeOnStreetParking	0/1 – whether on-street parking exists in scope
StreetInScopeParkingFormat	Parallel / Mixed / Angled / None (construction)
StreetInScopeOnStreetParkingCap	On-street parking capacity (spaces)
BufferOnStreetParking*	Buffer-zone parking attributes

Data quality

- Core structural fields (QUARTER, SiteID, STREET_SEGMENT_ID, capacity) are complete.
- Baseline, InterventionType, and InterventionSummary are sparsely populated (≈30,140 nulls) because most quarter-rows are simply “no change” periods.
- Parking format is missing on 877 rows; buffer fields have moderate missingness.

InterventionType distribution (non-null)

Type	Count
ProtectedBikeLane	377
Control	193
Regional Reallocation	61
Pedestrianisation	12

Parking capacity

- Overall mean capacity = 59 spaces (median 47, max 293).
- Baseline pre vs post for ProtectedBikeLane: Baseline mean = 50.8 spaces – Post mean = 36.7 spaces. The average reduction of roughly 14 spaces after protected bike lane interventions.
- Pedestrianisation rows show capacity dropping to 0 after intervention.

Parking format

- Parallel dominates (=24,000 rows), followed by Mixed and Angled.

3. Dataset 2 – Sites Register (sites_db)

Shape: 637 rows × 35 columns

Key columns

Column	Description
InterventionID / SiteID / StreetSegmentID	Identifiers
SiteType	Intervention or Control
Baseline	Yes / No
StreetInScope	Street name
DateOfIntervention/InitialCapture	Capture or intervention date
DisruptionStartDate / DisruptionEndDate	Construction window
InterventionType	ProtectedBikeLane, Control, Regional Reallocation, Pedestrianisation
StreetInScopeOnStreetParkingCapacity	On-street capacity (spaces)
StreetInScopeParkingFormat	Parallel / Mixed / Angled
Buffer*, OffStreet*, NumberOfTrafficLanes, PedestrianEstimatesAvgDay	Supporting attributes
InterventionSummary / Expanded / Comment	Free-text notes

SiteType & InterventionType

SiteType	Count
Intervention	442
Control	195

InterventionType	Count
ProtectedBikeLane	416
Control	195
Regional Reallocation	12
Pedestrianisation	9

Baseline flag

- Yes: 274 rows (pre / baseline observations)
- No: 363 rows (post or intermediate observations)

Parking capacity by SiteType

SiteType	Mean capacity	Median	Min	Max
Control	56.9	48	0	245
Intervention	38.6	26	0	293

Intervention sites show lower average on-street capacity than control sites, consistent with space reallocation.

Dominant streets are the LA TROBE STREET, Abbotsford St, St Kilda Rd, Albert Street, High Street, Southbank Boulevard, William Street, etc.

Notable patterns

- Many rows for the same SiteID / StreetSegmentID represent successive states (baseline, during works, post).
- Disruption dates are available for a subset of sites and should be used to exclude construction periods from utilisation analysis.
- Free-text Comments contain useful caveats (for example, temporary parking loss, tram-stop upgrades, level-crossing works).

4. Join & Analytical Compatibility

Link	Assessment
SiteID / STREET_SEGMENT_ID across the two files	Primary keys for linking quarterly panel to site register
StreetInScope / StreetName	Can be matched to sensor data StreetName after standardisation
STREET_SEGMENT_ID	Candidate for joining to street_spatial.gpkg
DateOfIntervention + Disruption windows	Define pre / during / post periods for DiD
StreetInScopeOnStreetParkingCap	Outcome / covariate for capacity-change analysis

Recommended matching sequence for sensor data

1. Standardise street names between sensor files and StreetInScope.
2. Restrict sensor events to the STREET_SEGMENT_IDs / streets present in these two files.
3. Use Baseline + DateOfIntervention + Disruption dates to assign each sensor event to pre / during / post.
4. Spatially validate against street_spatial.gpkg where attribute matches are ambiguous.

5. for the Research Question

These two datasets are central to the study design:

- They explicitly define treatment (mainly ProtectedBikeLane, plus Pedestrianisation and Regional Reallocation) versus control streets.
- They record on-street parking capacity before and after interventions, showing clear average reductions on treatment streets.
- The quarterly panel allows construction of a time-series of capacity and intervention status.
- Disruption date fields enable exclusion of construction periods.

Limitations

- Capacity is recorded as a point estimate per state; it is not continuous sensor occupancy.
- Street-name inconsistency (case, abbreviations) will require cleaning before joining to City of Melbourne sensor data.
- Some control sites still experience other works (level crossings, building construction); these are noted in Comments and should be handled carefully.
- Not every IV site will necessarily appear in the 2011–2020 sensor coverage because of geographic and temporal gaps remain.

Overall assessment together with the 2012 -2020 sensor event files, these datasets enable a Difference-in-Differences design: Treatment vs Control and Pre vs Post, using both capacity change and sensor-derived utilisation metrics.

6. Next Steps

1. Standardise street names across StreetInScope, quarterly descriptions, and sensor StreetName.
2. Build a clean lookup of SiteID / STREET_SEGMENT_ID to treatment/control flag and intervention date(s).
3. Calculate capacity change (baseline to post) by InterventionType and SiteID.
4. Filter 2012- 2020 sensor data to the streets present in these files and produce baseline utilisation metrics.
5. Define exclusion windows using DisruptionStartDate / DisruptionEndDate.